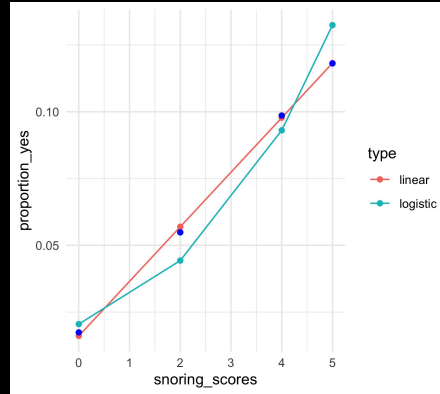


Generalized Linear Models



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Introduction to Public Health Informatics

DH 302

Lecture 13 || Wednesday, 19th February 2025

Sorry... And THANK YOU!

In a land where data charts unfold,
And p-values whisper tales untold,
A noble quest, a task was laid—
To test the means, the truth displayed.

The test of means was set with care,
But typos lurked in symbols there!
A sigma lost, a mu askew—
The code just broke, the errors grew.

Yet through the mess, they stood their ground,
Debugging till the truth was found.
For every typo, every plight,
Made their stats skills sharp and bright!

Co-authored with ChatGPT

Revisiting OLS

- In ordinary least squares (OLS or linear regression), we model the the “response” (dependent) variable y as a linear combination of the “explanatory” (independent) variables x
- The fit predicts the expected value of the “response” variables as a linear combination of the “explanatory variables
- The assumption of linear regression are usually good for variables where the variation in the mean of the response variable is small as compared to change in the explanatory variables
- However, the expected response can change by large amounts (say exponentially) with a small change in explanatory variables when the usual assumptions of OLS (which ones?) start to fail
- Example: Change in the level of snoring can increase the probability of developing a heart disease by 4x

Generalized linear models

- Linear models assume a linear relationship between the mean “response” variable (dependent) Y and the set of “covariates” or “independent” or “explanatory” variables.
 - Y is assumed to be normal
 - Models linear function of the mean
- Generalized linear models: Extend normal distributions to model non-normal response (dependent variable) as a function and allow for non-linear functions of the mean:
 - Response variable: Specifies the response variable and its probability distribution
 - Explanatory variables: Specifies the p explanatory variables for a linear predictor
$$\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}$$
 x_{ij} Value of explanatory variable j for observation i
 - Link function: A function that maps the mean of the response to the explanatory variables

$$\mu_i = E(Y_i \mid x_{i1}, \dots, x_{ip})$$

$$g(\mu_i) = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}.$$

Link function

- The probability distribution function captures how Y is distributed
- The linear predictor captures the dominance of independent factors
- The link function captures the relationship between linear predictor and the mean of the distribution function

Distribution	Support of distribution	Typical uses	Link name	Link function, $\mathbf{X}\beta = g(\mu)$	Mean function
Normal	real: $(-\infty, +\infty)$	Linear-response data	Identity	$\mathbf{X}\beta = \mu$	$\mu = \mathbf{X}\beta$
Exponential	real: $(0, +\infty)$	Exponential-response data, scale parameters	Negative inverse	$\mathbf{X}\beta = -\mu^{-1}$	$\mu = -(\mathbf{X}\beta)^{-1}$
Gamma					
Inverse Gaussian	real: $(0, +\infty)$		Inverse squared	$\mathbf{X}\beta = \mu^{-2}$	$\mu = (\mathbf{X}\beta)^{-1/2}$
Poisson	integer: $0, 1, 2, \dots$	count of occurrences in fixed amount of time/space	Log	$\mathbf{X}\beta = \ln(\mu)$	$\mu = \exp(\mathbf{X}\beta)$
Bernoulli	integer: $\{0, 1\}$	outcome of single yes/no occurrence	Logit	$\mathbf{X}\beta = \ln\left(\frac{\mu}{1 - \mu}\right)$	$\mu = \frac{\exp(\mathbf{X}\beta)}{1 + \exp(\mathbf{X}\beta)} = \frac{1}{1 + \exp(-\mathbf{X}\beta)}$
Binomial	integer: $0, 1, \dots, N$	count of # of "yes" occurrences out of N yes/no occurrences		$\mathbf{X}\beta = \ln\left(\frac{\mu}{n - \mu}\right)$	
Categorical	integer: $[0, K)$	outcome of single K-way occurrence		$\mathbf{X}\beta = \ln\left(\frac{\mu}{1 - \mu}\right)$	
	K-vector of integer: $[0, 1]$, where exactly one element in the vector has the value 1				
Multinomial	K-vector of integer: $[0, N]$	count of occurrences of different types (1, ..., K) out of N total K-way occurrences			Source

[Source](#)

Advantage of GLM rather than transforming

- Before GLMs, a common way to handle response variables that violate the assumptions of OLS was to transform the variable such that the transformation was approximately normal (log/square root/...)
- However finding the transformation that achieves this is itself tricky
- It would forcefully model $E[g(Y)]$ while we want to ideally model $E(Y)$
- GLMs solve this problem by allowing Y to have a flexible distribution $g[E(Y)]$

Logistic regression

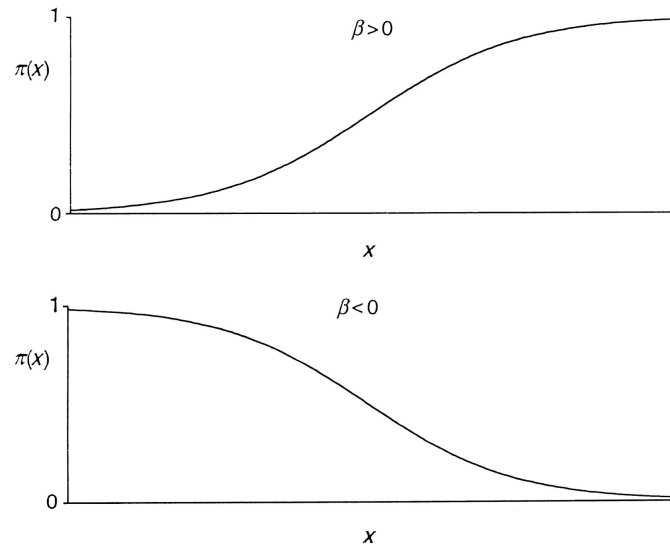
y_i takes value 0 or 1

$$\pi_i = \mu_i = E(Y_i) = P(Y_i = 1)$$

$$\text{logit}(\pi_i) = \log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}, \quad i = 1, \dots, n.$$

$$P(Y_i = 1) = \pi_i = \frac{\exp(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip})}{1 + \exp(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip})}.$$

$$\text{odds} = \frac{P(Y = 1)}{P(Y = 0)}. \quad \frac{P(Y = 1)}{P(Y = 0)} = \frac{\exp(\beta_0 + \beta_1 x) / [1 + \exp(\beta_0 + \beta_1 x)]}{1 / [1 + \exp(\beta_0 + \beta_1 x)]} = \exp(\beta_0 + \beta_1 x) = e^{\beta_0} (e^{\beta_1})^x$$



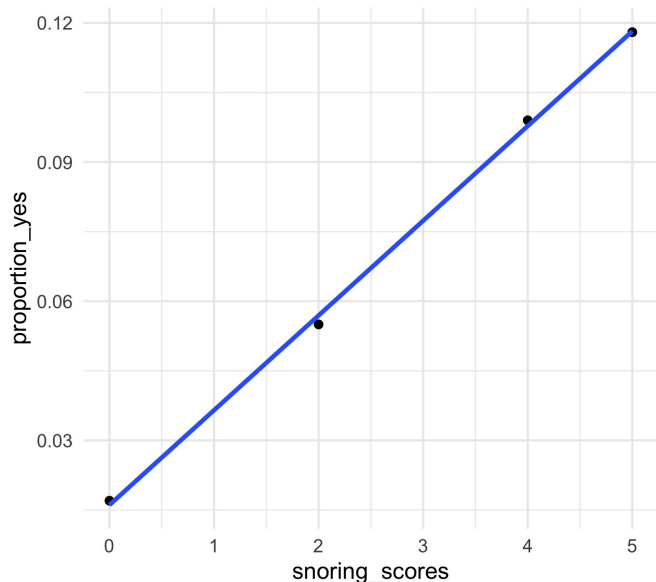
Example: Snoring and heart disease

Snoring	Heart Disease		Proportion Yes
	Yes	No	
Never	24	1355	0.017
Occasional	35	603	0.055
Nearly every night	21	192	0.099
Every night	30	224	0.118

[Source](#)

Scoring levels: 0 (Never), 2(occasional), 4(Nearly every night), 5(every night)

Snoring and heart disease



Call:

```
lm(formula = proportion_yes ~ snoring_scores, data = df)
```

Residuals:

1	2	3	4
0.0009153	-0.0019322	0.0012203	-0.0002034

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.0160847	0.0015254	10.54	0.008874 **
snoring_scores	0.0204237	0.0004548	44.91	0.000495 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.001747 on 2 degrees of freedom

Multiple R-squared: 0.999, Adjusted R-squared: 0.9985

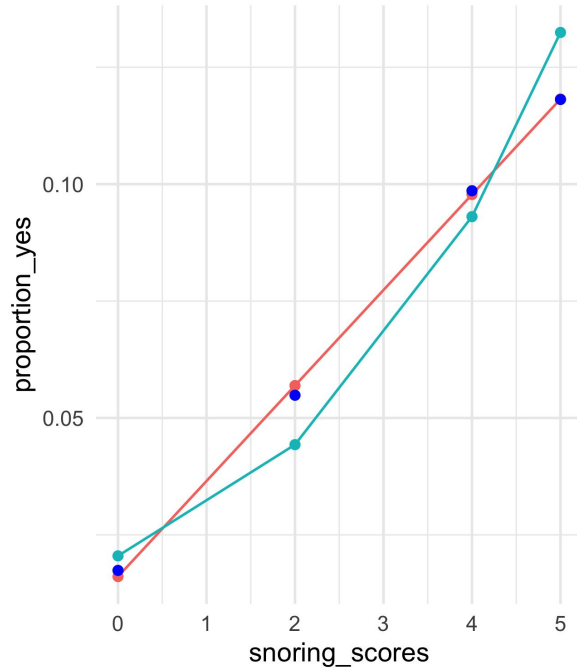
F-statistic: 2017 on 1 and 2 DF, p-value: 0.0004955

$$P(Y=1) = 0.016 + 0.02x$$

Model interpretation:

- Estimated probability is 0.016 for non-snorers
- Increases by $0.02 \times 2 = 0.04$ for occasional scores, another 0.04 for those who snore every night and another 0.02 for who always snore

Snoring and heart disease: Logistic model



Call:

```
glm(formula = proportion_yes ~ snoring_scores, family = binomial(link = "logit"),  
     data = df, weights = total)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.86625	0.16621	-23.261	< 2e-16 ***
snoring_scores	0.39734	0.05001	7.945	1.94e-15 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 65.9045 on 3 degrees of freedom

Residual deviance: 2.8089 on 2 degrees of freedom

AIC: 27.061

Generalized linear models for counts and rates

- Counts have positive means
- Poisson with loglinear model

$$\log \mu_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}, \quad i = 1, \dots, n.$$

$$\mu_i = \exp(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip})$$

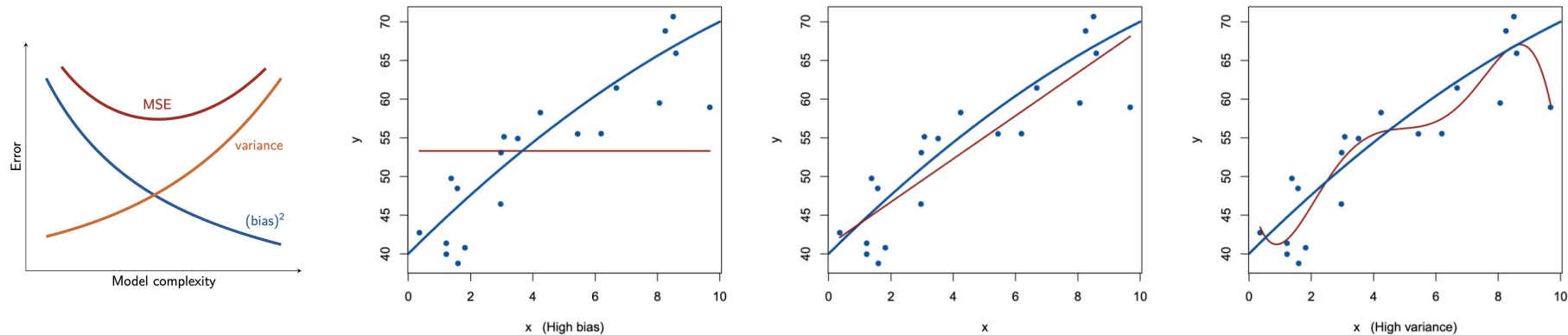
Counts: Crabshoe example

- During spawning season, a female migrates to the shore to breed. With a male attached to her posterior spine, she burrows into the sand and lays clusters of eggs
- Eggs are fertilized externally, in the sand beneath the pair
- During spawning other male crabs, called satellites, may cluster around the pair and also fertilize the eggs
- Response outcome for each female crab is y = number of satellites
- Explanatory variables:
 - female crab's color (1 = medium light; 2 = medium; 3 = medium dark; 4 = dark) (Color is a surrogate for the age of the crab, as older crabs tend to have a darker color.)
 - spine condition (1 = both good; 2 = one worn or broken; 3 = both worn or broken)
 - carapace width (cm), and weight (kg).

sat	color	spine	width	weight	sat	color	spine	width	weight
8	2	3	28.3	3.05	0	3	3	22.5	1.55
4	3	3	26.0	2.60	0	2	3	23.8	2.10
0	3	3	25.6	2.15	0	3	3	24.3	2.15
0	4	2	21.0	1.85	14	2	1	26.0	2.30



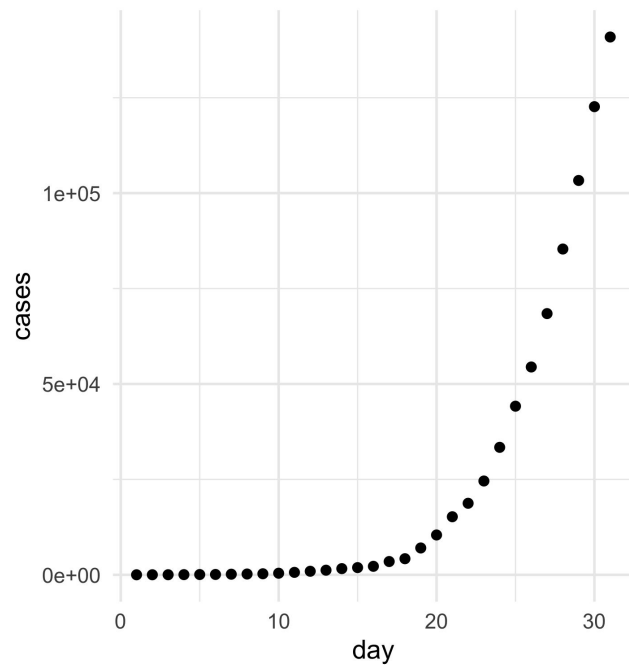
Assessing Model Fit



AIC judges a model by how close its fitted values tend to be to the true expected values, as summarized by a certain expected distance between the two

$$\text{AIC} = -2(\log \text{likelihood}) + 2(\text{number of parameters in model}) :$$

Covid Example



Questions?

